ORBTRAN-II User Manual



Wenli Zou
Institute of Modern Physics, Northwest University,
Xi'an, Shaanxi, China

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1 About OrbTran-II

OrbTran-II is a Fortran program designed to convert pseudo-canonical molecular orbital (PCMO) data files – generated by the quadratically convergent state-averaged MCSCF-type (CASSCF, RASSCF, or GASSCF) calculations using Molpro or BDF – into formats compatible with other programs, most notably OpenMolcas. This conversion facilitates the comparison of results obtained from multi-reference methods such as MRCI, CASPT2, NEVPT2, and MCPDFT by different quantum chemistry programs.

The OrbTran-II program is an upgraded version of the original OrbTran tool, which was first developed in 2020. OrbTran was designed to convert PCMO files from Molpro or BDF into the RasOrb format used by OpenMolcas calculations. At the time, similar tools already existed, such as MOKIT (`https://gitlab.com/jxzou/mokit`), but OrbTran was built specifically for MCSCF calculations involving heteronuclear diatomic molecules with C_2 or C_{2v} symmetry, as well as atoms with C_{2v} or D_{2h} symmetry. This specialization, however, limited its use for general molecular systems. OrbTran-II is the successor to OrbTran, extending support to general molecules.

The OrbTran-II program has been applied in theoretical researches on electronic spectra of some diatomic molecules. See the following references.

1. D. He, C. Li, and W. Zou, *Theoretical study on the low-lying electronic states of ScO*, **J. Quant. Spectrosc. Radiat. Transf.** 345, 109559 (2025). `https://doi.org/10.1016/j.jqsrt.2025.109559`

2. C. Li and W. Zou, *Low-lying electronic states of RhF and IrF: A theoretical study*, **Chem. Phys. Lett.** 882, 142454 (2026). `https://doi.org/10.1016/j.cplett.2025.142454`

3. C. Li and W. Zou, *Electronic states of cobalt monofluoride (CoF) below 5 eV*, **J. Quant. Spectrosc. Radiat. Transf.** 357, 109910 (2026). `https://doi.org/10.1016/j.jqsrt.2026.109910`

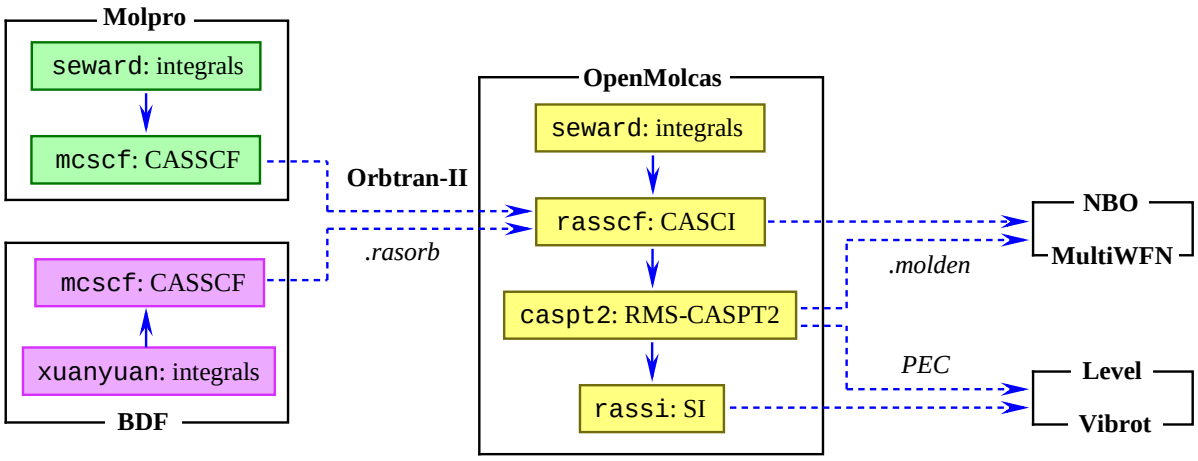


Figure 1: Flowchart of OrbTran-II program transferring MCSCF orbital files from Molpro and BDF to OpenMolcas.

1.1 Features

OrbTran-II provides the following features.

1. Reads PCMOs.

    - PCMOs may be computed by Molpro's `MULTI` module or BDF's `MCSCF` module.
    - All eight Abelian point groups have been supported.
    - Spherical Gaussian functions with angular momenta ranging from 0 to 6 (*spdfghi*) can be identified. To convert to the MOLEKEL/GAMESS/MOLDEN format, the basis functions must be written to the output file in a specific format. See the examples.

2. Writes PCMOs.

    - RasOrb format used by BDF (`MCSCF`), Molpro (`MATROP` & `MULTI`), or OpenMolcas (`RASSCF`).
    - mocoef format used by Columbus, accessed via the `MCSCF` module using `Argos` integrals up to *g*-functions. Symmetry-equivalent atoms are not supported.
    - MOLEKEL format used by ORCA (up to *h*-functions). Symmetry-equivalent atoms are not supported [1].
    - PUNCH format used by GAMESS and Firefly (PC-GAMESS). For GAMESS before 2013 and Firefly, only *spdfg*-functions are supported. Symmetry-equivalent atoms are not supported.
    - MOLDEN format for display and plotting (up to *h*-functions). Symmetry-equivalent atoms are not supported.
    - A new RasOrb file with adjusted PCMO coefficients.
    - To do: data formats used by PySCF, ...

Figure 1 shows the flowchart of using the OrbTran-II program to prepare SA-CASSCF molecular orbitals, from either Molpro or BDF, for RMS-CASPT2 + SOC calculations by OpenMolcas. Subsequent utility programs then perform data processing and analysis.

[1] In principle, MOLDEN files generated by Molpro's or BDF's MCSCF calculations can be converted to the MOLEKEL format directly, but I have not found a tool that performs this conversion; perhaps IOData (https://github.com/theochem/iodata) can do this.

2 Building and Running OrbTran-II

A Fortran90 compiler is required to compile OrbTran-II, *e.g.* `gfortran`:

```
$ gfortran -O3 orbtran2.f90 -o orbtran2.exe
```

To run OrbTran-II, you need to specify two files generated by Molpro (or BDF) MCSCF calculations, the standard output file and the RasOrb data file, as well as the wanted format(s) to be saved. For example, the following command performs the conversion from Molpro to OpenMolcas and BDF

```
$ ./orbtran2.exe -log 01.out -mos 01.casorb -ras 02.rasorb -bdf 03.rasorb
```

For an explanation of these options, see Section 3.

You can also create a batch file, using the sample text provided below, to perform conversion on a series of RasOrb data files.

bash script

```
#!/bin/bash

CurrDir=$(pwd)
EXE="$CurrDir/tools/orbtran2.exe -n 1"
MODIR=$CurrDir/H2O/CASSCF
OUT=-out:molcas
SVDIR=$CurrDir/Molcas/H2O

mkdir -p $SVDIR

DIST=1.80;  $EXE -LOG $MODIR/$DIST.out -MOS $MODIR/$DIST.rasorb $OUT $SVDIR/$DIST.rasorb
DIST=1.82;  $EXE -LOG $MODIR/$DIST.out -MOS $MODIR/$DIST.rasorb $OUT $SVDIR/$DIST.rasorb
DIST=1.84;  $EXE -LOG $MODIR/$DIST.out -MOS $MODIR/$DIST.rasorb $OUT $SVDIR/$DIST.rasorb
```

3 Description of Control Options

All options in the OrbTran-II program are case-insensitive, with the details as follows.

Option	Abbr.	Description
Options for Controlling Input		
`-PRO` *name*		This option specifies which program generated the MCSCF results; *name* can be `Molpro` or `BDF`. By default, *name* is determined by the PCMO data file (specified by the `-MOS` option; see below), so this option generally does not need to be specified.
`-LOG` *file*		This option specifies Molpro's or BDF's MCSCF output file. If not provided, the default file name is `mcscf.out` .
`-N` *integer*		In the output file of a multi-step MCSCF task, this option specifies which step's MCSCF output to read. If not provided, the outputs of the first MCSCF calculation will be read.
`-MOS` *file*		This option specifies Molpro's or BDF's RasOrb data file. If not provided, the default file name is `mcscf.casorb` .
`-OPR` *file*		This option specifies a file for orbital operations; see below for details. The modified RasOrb file will be saved in the file specified by the `-NEW` option (see below).
Options for Controlling Output		
`-OUT:BDF` *file*	`-BDF` *file*	This option specifies BDF's RasOrb data file.
`-OUT:ARGOS` *file*	`-ARG` *file*	This option specifies Columbus' mocoef data file, read by the `MCSCF` module with `Argos` integrals.
`-OUT:MOLCAS` *file*	`-RAS` *file*	This option specifies Molcas's RasOrb data file.
`-OUT:MOLPRO` *file*	`-MOL` *file*	This option specifies MOLPRO's RasOrb data file.
`-OUT:MOLEKEL` *file*	`-MKL` *file*	This option specifies a MOLEKEL data file, which can be converted into a formatted `.gbw` file using the `orca_2mkl` tool, and subsequently used in ORCA calculations.
`-OUT:GAMESS` *file*	`-GAM` *file*	This option specifies the PUNCH data file used by Firefly (PC-GAMESS) and GAMESS.
`-OUT:MOLDEN` *file*	`-MDN` *file*	This option specifies a MOLDEN data file.
`-OUT:NEW` *file*	`-NEW` *file*	This option specifies a new RasOrb data file, in the same format as the old one, used to save the modified PCMOs (see `-OPR` above). If this option is not specified, but `-OPR` is specified, the modified PCMOs will be saved to `new_mcscf.casorb` .

The text file read by the `-OPR` option consists of a series of PCMO operations (case-sensitive), with one operation executed per line, as follows.

Operation	Description
`x` *isym imo1 imo2*	E[x]change PCMO arrays *imo1.isym* and *imo2.isym*.

`c` *isym imo*	[c]lean the PCMO array *imo.isym*, *i.e.* zero out the array.
`p` *isym imo*	Change the [p]hase of PCMO array *imo.isym*, *i.e.* multiply the array by -1.
`P` *isym imo*	[P]rint the coefficients of PCMO *imo.isym* to the screen.
`s` *isym imo ic val*	[s]et the *ic*-th coefficient in PCMO array *imo.isym* to *val*.
`S` *isym imo ic val*	It is equivalent to the two operations `c` + `s`; clean the PCMO array *imo.isym*, and [S]et the *ic*-th coefficient to *val*.

4 Input Templates and Required Keywords

To generate the data required by the OrbTran-II program and ensure it is read correctly, some specific keywords must appear in the input file of Molpro or BDF. The complete input files can be found in the `example` directory.

4.1 Molpro

Below is a Molpro input illustrating the SA-CASSCF(10e,8o) calculation of the ground state $X\,^1\Sigma^+$ and two $^3\Pi$ excited states of the BF molecule, using the scalar X2C relativistic Hamiltonian with the finite nuclear model. The example performs two `multi` runs; the second run produces the RasOrb file, so the `-n 2` option should be specified for the OrbTran-II program [2].

Input of Molpro

```
***,BF
file,2,bf.wf
memory,200,m
gthresh,thrprint=0                                  ! Required.
gprint,basis=1                                      ! Required for -MKL, -MDN, and -GAM.

! ANO-R1 (truncated)
basis={
S,F ,0.785982500E+05,0.119374900E+05,...
...

S,B ,0.104312300E+05,0.161792600E+04,...
...
}

! sf-X2C, with the finite nuclear model
set,fnuc=1,dkho=101

geometry={
B   0.0  0.0  1.4
F   0.0  0.0  0.0}

$dist=140
$fname=bf-$dist.casorb

! RHF
{hf;occ,5,1,1,0;wf,14,1,0;}

! SS-CASSCF(10e,8o)
{multi;
start,2100.2;orbital,2140.2;
frozen,0;closed,2,0,0,0;occ,6,2,2,0;
wf,14,1,0;state,1; ! X^1Sigma^+
}

! SA-CASSCF(10e,8o)
```

[2] This case is special: since the first `multi` run does not print molecular orbitals, omitting `-n 2` will not cause errors, but this is generally not recommended.

```
{multi;
canorb,ci;orbprint,nvirt=1;                          ! Required.
start,2140.2;orbital,2141.2;
frozen,0;closed,2,0,0,0;occ,6,2,2,0;
wf,14,1,0;state,1; ! X^1Sigma^+
wf,14,2,2;state,2; ! a^3Pi, b^3Pi
wf,14,3,2;state,2;
}

{matrop                                              ! Required to save a RasOrb file.
load,ccan,orb,2141.2,canonical;
write,ccan,$fname,new,scientific;
}
---
```

In the input above, the `matrop` module is used to save a Rasorb file. The `matrop` module can also read RasOrb files saved by either itself or the OrbTran-II program through `-MOL`. Use the input below (the content before molecular coordinates has been omitted):

Input of Molpro

```
...
geometry={
B    0.0   0.0   1.4
F    0.0   0.0   0.0}

$fname=new_mcscf.casorb

{matrop
read,ccan,file=$fname;
save,ccan,2150.2,orbitals;
}

! SA-CASSCF(10e,8o)
{multi;
canorb,ci;orbprint,nvirt=1;
start,2150.2;orbital,2151.2;
frozen,0;closed,2,0,0,0;occ,6,2,2,0;
wf,14,1,0;state,1; ! X^1Sigma^+
wf,14,2,2;state,2; ! a^3Pi, b^3Pi
wf,14,3,2;state,2;
}
---
```

4.2 BDF

The RasOrb file generated by the above Molpro calculation, after conversion by the OrbTran-II program (for example, the file named `bf-140.bdforb`), can be read using the following BDF input.

Input of BDF

```
$compass
basis
 ano-r1              # ensure the basis functions are consistent with those in Molpro.
```

```
geometry
 B   0.0  0.0  1.4    # ensure coordinates are consistent.
 F   0.0  0.0  0.0
end geometry
group
 gen(x,y)             # always use gen() to specify the Abelian symmetry (C2v here).
saorb
$end

$xuanyuan             # sf-X2C, with the finite nuclear model.
heff
 3
clight                # use the built-in speed of light in Molpro/Molcas.
 2
nuclear
 1
$end

%cp /home/zouwl/Work/Molpro/bf-140.bdforb $BDF_WORKDIR/$BDFTASK.inporb

$mcscf
casci                 # CASCI does not optimize orbitals.
core
   0  0  0  0         # The irreps. of C2v are a1, a2, b1, b2, unlike in Molpro.
close
   2  0  0  0
active
   4  0  2  2
actel
  10
mixci
  3                   # average three states: X^1Sigma^+ and two 3Pi states.
   1  3  3
   1  2  2
   1  3  4
guess
 read                 # read initial orbitals from $BDFTASK.inporb.
quasi
nograd
$end
```

For BDF, you need to use the `gen()` function after the `GROUP` keyword to specify Abelian symmetry generators. It is not recommended to directly specify the point group name (unless you are familiar with the details of molecular coordinates, symmetry, and basis contractions), as this may cause unexpected rotation of molecular coordinates. For sf-X2C relativistic calculations, it may also be necessary to modify the nuclear charge distribution model (`NUCLEAR` or `NUCLEARRADII`) and the speed of light (`CLIGHT`) in the `XUANYUAN` module in order to be consistent with the settings of Molpro.

When performing MCSCF calculations with BDF and generating PCMO data,

1. specify `Iprtmo = 2` in the `MCSCF` module. This function is analogous to Molpro's `orbprint`.

2. to output MOLEKEL or MOLDEN format, the printing format of basis functions should also be adjusted by setting `Expbas = ORCA` or 4 in the `COMPASS` module.

Please refer to the example in `/example/bdf-to-others-ecp/bdf` . Upon completion of the BDF MCSCF calculation, a RasOrb file is automatically generated and saved in the scratch directory as `$BDFTASK.casorb`.

4.3 OpenMolcas

The OpenMolcas input used to read the transformed RasOrb file via the `LUMORB` keyword is shown below. The `RASSCF` module cannot average states with different spins or different *irreps.*, so a SA-CASSCF calculation has to be split into multiple CASCI (the `CIONLY` keyword) calculations without orbital optimization, which may exactly reproduce Molpro's SA-CASSCF results. Because OpenMolcas can occasionally lose roots, especially when electronic states are densely distributed, you can slightly increase the number of averaged roots specified by `CIROOT`.

Input of OpenMolcas

```
&GATEWAY
  Symmetry = X Y
  RX2C
  Finite
  Basis set
    B.ANO-R...3s2p1d.
    B     0.000000    0.000000    1.400000    Angstrom
  End of basis
  Basis set
    F.ANO-R...3s2p1d.
    F     0.000000    0.000000    0.000000    Angstrom
  End of basis

>>EXPORT DIST=140
>>EXPORT ORB_FILE=/home/zouwl/Work/Molpro/bf-${DIST}.rasorb

&SEWARD

*================================================================================ 1A1
>>COPY $ORB_FILE $Project.RasOrb
&RASSCF
Spin      = 1
Symmetry  = 1
CIROOT    =  1  1  1
Frozen    =  0  0  0  0
Inactive  =  2  0  0  0
Ras2      =  4  2  2  0
NActEl    = 10  0  0
Lumorb
OUTOrbitals = CANOnical
CIONLY
End of input

*================================================================================ 3B1
>>COPY $ORB_FILE $Project.RasOrb
&RASSCF
```

```
Spin      = 3
Symmetry  = 2
CIROOT    =  2  2  1
Frozen    =  0  0  0  0
Inactive  =  2  0  0  0
Ras2      =  4  2  2  0
NActEl    = 10  0  0
Lumorb
OUTOrbitals = CANOnical
CIONLY
End of input

*================================================================================ 3B2
>>COPY $ORB_FILE $Project.RasOrb
&RASSCF
Spin      = 3
Symmetry  = 3
CIROOT    =  2  2  1
Frozen    =  0  0  0  0
Inactive  =  2  0  0  0
Ras2      =  4  2  2  0
NActEl    = 10  0  0
Lumorb
OUTOrbitals = CANOnical
CIONLY
End of input
```

For transition-metal systems with densely distributed electronic states, root loss may occur. In this case, slightly increase the number of averaged roots in the `CIROOT` keyword, typically by 1−2. Since CASSCI does not optimize orbitals, the lower-energy roots will not change.

Using ECP basis sets that are not included in OpenMolcas basis set library is not straightforward. Please refer to the example in `/example/bdf-to-others-ecp/molcas` and follow the instructions below.
https://www.molcas.org/molcas/wwwboard/messages/1531.html
https://molcas.gitlab.io/OpenMolcas/sphinx/users.guide/basis_library.html#structure-of-the-ecp-libraries

4.4 ORCA

The MOLEKEL file produced by the OrbTran-II program (for example, `bf-140.mkl`) can be converted into a formatted `.gbw` file with ORCA's `orca_2mkl` tool

```
$ {orca_path}/orca_2mkl bf-140 -gbw
```

The `.gbw` file can be read using the `MORead` keyword of ORCA. Below is an ORCA input in which the basis functions have been truncated. The complete input is available in the `example` directory.

Input of ORCA

```
# sf-X2C/CASSCF/ANO-R1

```

```
! PAL4 VeryTightSCF UseSym X2C Moread

%Moinp "/home/zouwl/Work/Orca/bf-140.gbw"

%MaxCore 1000

%basis
  NewGTO  B
S    12
  1       10431.23000000      0.00030668
  2        1617.92600000      0.00197068
  3         353.05470000      0.01049802
(omitted)
  end
  NewGTO  F
S    13
  1       78598.25000000      0.00017799
  2       11937.49000000      0.00090402
  3        2534.68900000      0.00443005
(omitted)
  end
end

%rel
  FiniteNuc true
  Velocity  137.0359991766    # clight used in openmolcas and molpro
end

%casscf nel     10
        norb    8
        mult    1, 3, 3
        irrep   0, 2, 3  # a1, b1, b2; note: numbering starts from zero!
        nroots  1, 2, 2
        orbstep nr
        maxiter 1        # casci
end

* xyz 0 1
 B   0.0  0.0  1.4
 F   0.0  0.0  0.0
*
```

4.5 Columbus

After generating the MCSCF input file with the `colinp` utility, open the `mcscfin` file and confirm that the `npath` parameter array contains 13 (for example, `npath=...,13,...`, which should have been set by default). Then rename the converted data file to `mocoef` and place it in the `WORK` directory (if you're running Columbus for the first time and don't know where the `WORK` directory is, perform an SCF calculation first).

4.6 GAMESS & Firefly

In most cases, specify `GUESS=MOREAD` and `NORB` in the `$GUESS` group. Then copy the converted orbital data, along with `$VEC` and `$END`, to the end of the input file.

For calculations with `RUNTYP=TRANSITN` (*e.g.*, transition moment and spin-orbit coupling calculations), the `$GUESS` group is not needed because of `SCFTYP=NONE`. Simply change `$VEC` to `$VEC1`.

4.7 Reference Energies

Table 1 lists the energies of BF calculated by five programs, demonstrating that Molpro's X2C(f.n.)/SA-CASSCF(10e,8o) results are exactly (or approximately) reproduced by other programs.

Table 1: BF at 1.4 Å.

Program	$X^1\Sigma^+$	$a^3\Pi$	$b^3\Pi$
Molpro	-124.29720532	-124.17354376	-123.87136154
BDF	-124.29720533	-124.17354376	-123.87136154
OpenMolcas [a]	-124.29720532	-124.17354376	-123.87136154
ORCA [a]	-124.29720532	-124.17354376	-123.87136154
GAMESS [a,b]	-124.29701826	-124.17335680	-123.87117428

[a] By CASCI.
[b] IOTC(p.n.) relativistic Hamiltonian.

Table 2 presents the ground-state $X^2\Pi$ and excited-state $a^4\Sigma^-$ energies of the SnI molecule, calculated at the SA-CASSCF(11e,8o)/DHF-SVP level using the BDF program, alongside the corresponding SA-CASSCF or CASCI results obtained from other programs.

Table 2: SnI at 2.7 Å.

Program	$X^2\Pi$	$a^4\Sigma^-$
BDF	-508.06875848	-507.98347986
Molpro	-508.06875815	-507.98348052
OpenMolcas [a]	-508.06875848	-507.98347987
ORCA [a]	-508.06875848	-507.98347986
Firefly [a]	-508.06875849	-507.98347988
Columbus	-508.06875792	-507.98348003

[a] By CASCI.

Example input and output files can be found in the `example` directory.

5 Notes, Limitations, and Known Issues

- The atoms computed by different programs must be in the same ordering and the same orientation.

  For instance, some programs may rotate molecular coordinates by default, which can lead to inconsistencies between the coordinate systems used by two different programs and cause the converted PCMOs to become invalid.

  For BDF, it is recommended to use the `gen()` function to specify symmetry generators, thereby avoiding unwanted coordinate rotation.

- The basis sets used in the two programs must be exactly identical. This means that both the ordering of contractions and the contraction coefficients match precisely. Before performing any calculations, it is essential to **carefully compare the basis functions employed by the two programs for the atoms involved!**

  For example, the cc-pVQZ basis set in the BDF basis set library adopts the Gaussian-style of contraction for compatibility with older versions, resulting in contraction coefficients that differ from those of the standard basis set, and therefore BDF's cc-pVQZ should be regarded as a different basis set. The following compares the contraction coefficients of the *s*-functions for the F atom:

contraction coefficients of F

```
Standard-style                                  Gaussian-style
 0.000095 -0.000022  0.0  0.0  0.0               0.000095 -0.000022  0.0  0.0  0.0
 0.000738 -0.000172  0.0  0.0  0.0               0.000738 -0.000172  0.0  0.0  0.0
 0.003858 -0.000891  0.0  0.0  0.0               0.003858 -0.000891  0.0  0.0  0.0
 0.015926 -0.003748  0.0  0.0  0.0               0.015926 -0.003748  0.0  0.0  0.0
 0.054289 -0.012862  0.0  0.0  0.0               0.054289 -0.012862  0.0  0.0  0.0
 0.149513 -0.038061  0.0  0.0  0.0               0.149513 -0.038061  0.0  0.0  0.0
 0.308252 -0.086239  0.0  0.0  0.0               0.308252 -0.086239  0.0  0.0  0.0
 0.394853 -0.155865  0.0  0.0  0.0               0.394853 -0.155865  0.0  0.0  0.0
 0.211031 -0.110914  0.0  0.0  0.0               0.211031 -0.110914  0.0  0.0  0.0
 0.017151  0.298761  1.0  0.0  0.0               0.000000  0.000000  1.0  0.0  0.0
-0.002015  0.585013  0.0  1.0  0.0               0.000000  0.000000  0.0  1.0  0.0
 0.000869  0.271159  0.0  0.0  1.0               0.000000  0.000000  0.0  0.0  1.0
```

  In addition, the basis set libraries of some programs may change the order of these columns.

  In cases where the basis data are inconsistent, explicitly write the adjusted basis functions in the input file (for example, as shown in the input files in the `example` directory), or create a new basis set file for use.

- Symmetry-equivalent atoms are not supported for saving Molekel and Molden files. You need to lower the point group symmetry used in Molpro or BDF calculations. In addition, the MCSCF calculations of Molpro or BDF can also save Molden files with the highest Abelian symmetry.

- For ORCA, some group symmetries may be not correctly supported (depend on the specific molecules) since symmetry generators and orientations cannot be effectively controlled. When you encounter this issue, the most effective solution is to adjust the molecular coordinates and symmetry generators in Molpro or BDF.